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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	**HTL**
Selected Domain	**Domain 2 - ALPHA RED**
Selected SDG Track (1–17)	**SDG 8 – Decent Work and Economic Growth**
Project / Solution Name	**Quora Chat**
Name (Leader)	**Tng Kar Ming**
Email & Phone (Leader)	**karming0425@gmail.com / +60173553022**
University Affiliation(s)	**Asia Pacific University of Technology & Innovation**
Number of Team Members	**5**

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Micro, small and medium enterprises (MSMEs) play a significant role in Malaysia's economy, representing 96.1% of business establishments in 2024 (OECD, 2026) and contributing 39.7% of Malaysia's GDP in 2025 (DOSM, 2026). However, many small businesses operate with limited manpower while managing increasing customer enquiries and daily operations.

A single employee may handle multiple customer conversations simultaneously, often across English, Bahasa Melayu, and Chinese. Gathering requirements, preparing quotations, obtaining approvals, and following up with customers require significant manual effort. As conversations become longer, important details may be overlooked, resulting in incomplete requirements, inaccurate quotations, delayed responses, and lost business opportunities.

The challenge extends beyond customer communication. Once a quotation is accepted, employees must transfer the agreed requirements to operational teams, where fragmented information can cause repeated clarification and misunderstandings. Unsigned quotations may also be forgotten when employees prioritize other responsibilities.

Although customer relationship management and automation platforms exist, adopting multiple disconnected systems can introduce additional complexity for resource-constrained SMEs.

Therefore, the core problem is **how SMEs can efficiently manage growing customer enquiries while preserving accurate information and coordination throughout the customer lifecycle without proportionally increasing manpower**.

This challenge aligns with SDG 8, particularly Targets 8.2 and 8.3, by supporting productivity, technological innovation, and sustainable SME growth.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Quora Chat is an AI-assisted customer enquiry and quotation workflow designed as a reusable module within the Orbyte SME ecosystem. It connects a customer-facing conversational assistant with a company-facing workspace to coordinate customer requests from initial enquiry to operational handover.

The solution consists of four interconnected modules:

1. Knowledge Base: Businesses maintain public product information and internal operating procedures. AI assistants retrieve relevant knowledge based on their permitted access, helping provide consistent and context-appropriate information.

2. Quotation Inbox: A multilingual AI assistant communicates with customers, identifies their requirements, asks clarifying questions, and generates structured requirement briefs. Employees can monitor customer activity, review extracted information, resolve exceptions, and intervene when necessary.

3. Quotes: The system prepares quotation drafts using predefined pricing rules or staff-provided estimates. Authorized employees review, approve, reject, or revise quotations before customers receive generated PDF documents and eSign links. Automated follow-ups help businesses manage unsigned quotations.

4. Team Chat: Once a quotation is signed, the system creates an operational job within the simulated Orbyte Work environment. The responsible worker receives a structured handover containing confirmed requirements, agreed scope, and customer commitments. An internal AI assistant supports team collaboration by retrieving relevant information and identifying outstanding tasks.

The workflow also supports milestone-based invoices, simulated customer payments, and receipts, extending coordination from enquiry to job completion.

Quora Chat integrates generative AI APIs, PostgreSQL, deterministic business rules, and Replit cloud deployment to deliver a functional prototype.

By reducing manual coordination and preserving information across customer, sales, and operational interactions, the solution aims to help SMEs **quote faster, follow up consistently, and serve more customers with existing manpower**.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Component	Technology	Purpose
Frontend	TanStack Start, React, TypeScript	Customer and company-facing interfaces
Styling/UI	Tailwind CSS, shadcn/ui	Responsive and accessible user interfaces
API	Hono, oRPC	HTTP services and type-safe communication
Database	PostgreSQL, Drizzle ORM	Persistent customer, quotation, workflow, and audit records
Generative AI	Replit-managed OpenAI / OpenRouter	Multilingual conversations, requirement extraction, summarization, and drafting
Business Logic	TypeScript	Deterministic pricing, validation, approval rules, and workflow transitions
Background Processing	PostgreSQL-backed worker	Handover processing, retries, and asynchronous workflow tasks
Document Generation	React PDF	Quotation, agreement, and billing PDF generation
Hosting	Replit	Deployment of the functional application
Version Control	GitHub	Source code management and collaboration

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Our team adopted a problem-driven design-thinking approach by examining how limited manpower affects customer communication, quotation preparation, and internal coordination in SMEs.

Empathize and Define: We identified common operational difficulties, including repetitive requirement clarification, fragmented customer information, delayed approvals, and inconsistent handovers.

Ideate: Rather than developing a standalone chatbot, we designed an integrated workflow connecting customers, AI assistants, employees, and managers through the existing Orbyte ecosystem concept.

Prototype: We developed customer-facing and company-facing interfaces supported by configurable business knowledge, conversational requirement extraction, quotation management, eSign, and job handover. Five synthetic SME scenarios demonstrate how the workflow can be adapted across industries.

Test: We implemented automated unit and integration tests covering requirement consistency, quotation approval, concurrency, signing, worker recovery, and knowledge-access boundaries. Further usability and real-model evaluations are planned.

Deployment: The application is hosted on Replit using PostgreSQL and AI API integration. Before Phase 2, our priorities are validating the complete deployed workflow, improving user experience, benchmarking operational benefits, and resolving outstanding quality issues.

This iterative approach prioritizes reliability, ease of adoption, and practical SME value over unnecessary technical complexity.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	Your Response
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	**SDG 8.2:** Improve economic productivity through technological upgrading and innovation. **SDG 8.3:** Support entrepreneurship, productive activities, and the growth of micro and small enterprises. Quora Chat contributes by automating repetitive quotation tasks and improving SME operational efficiency.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	**20 minutes of active employee time per completed quotation**, obtained from a controlled manual-workflow evaluation using synthetic customer enquiries. The benchmark includes requirements gathering, quotation preparation, approvals, and operational handover.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	**Percentage reduction in active employee handling time per completed enquiry**, from initial requirements gathering through quotation approval and handover. Secondary metrics: requirement completeness (%) and handover information accuracy (%).
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	**Target: 30% reduction** in active employee handling time compared with the manual baseline, while maintaining requirement completeness and approval accuracy. This is a proposed prototype evaluation target, not an achieved result.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
React	Code framework	MIT
TanStack Start / Router	Code framework	MIT
Hono	Code framework	MIT
oRPC	Code framework	MIT
Drizzle ORM	Code framework	Apache License 2.0
PostgreSQL	Database	PostgreSQL License
Tailwind CSS	UI framework	MIT
shadcn/ui and underlying components	UI code/components	Verify applicable component licenses
React PDF	PDF generation library	MIT
OpenAI GPT-5 mini	Pre-trained AI model/API	Proprietary API/service terms
Replit	Cloud development and hosting platform	Commercial service terms
OpenRouter	AI API platform, local development	Platform service terms

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

Tng Kar Ming

Tng

24 October 2026